5-Bay SATA Drive Power Supply

Two 12V bricks (J8 -> ports 1-3, J9 -> ports 4-5), five analog-staggered channels, per-port 5V buck

5V rails: U2 -> ports 1,2,3 | U4 -> ports 4,5

Input protection



File: sata_psu_input.kicad_sch

Channel 1



File: sata_psu_channel1.kicad_sch

Channel 4



File: sata_psu_channel4.kicad_sch

5V rail links



File: sata_psu_rails.kicad_sch

Channel 2



File: sata_psu_channel2.kicad_sch

Channel 5



File: sata_psu_channel5.kicad_sch

Channel 3



File: sata_psu_channel3.kicad_sch

5-Bay SATA Drive PSU

Sheet: /
File: sata_psu.kicad_sch

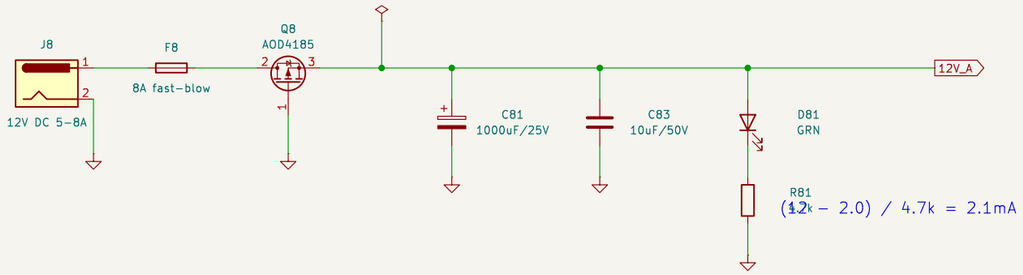
Title: 5-Bay SATA Drive Power Supply

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.3		Id: 1/8

Input protection — two 12V bricks, one per domain: fuse, reverse-polarity guard, bulk C

12V_A and 12V_B are never joined: bricks in parallel do not share, and an unplugged one would be back-fed. GND is common.

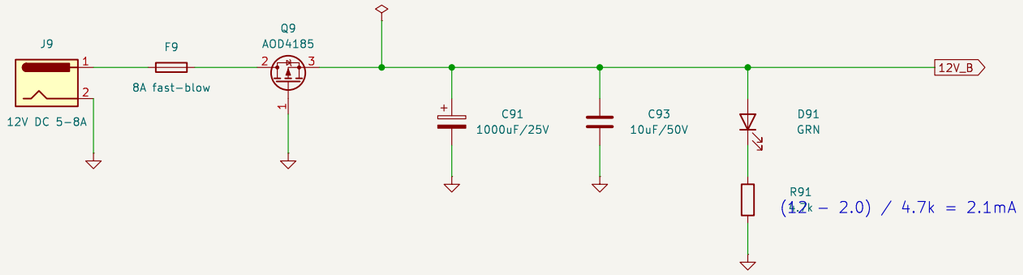
Input A — J8 feeds 12V_A, ports 1, 2, 3



(12 − 2.0) / 4.7k = 2.1mA

Star ground: single point back to both brick negatives

Input B — J9 feeds 12V_B, ports 4, 5



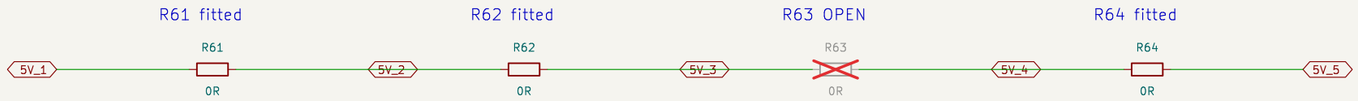
(12 − 2.0) / 4.7k = 2.1mA

5-Bay SATA Drive PSU		
Sheet: /Input protection/ File: sata_psu_input.kicad_sch		
Title: Input protection		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.3		Id: 2/8

5V rail links — how each port gets its 5V

group 1: U2 feeds ports 1, 2, 3
group 2: U4 feeds ports 4, 5

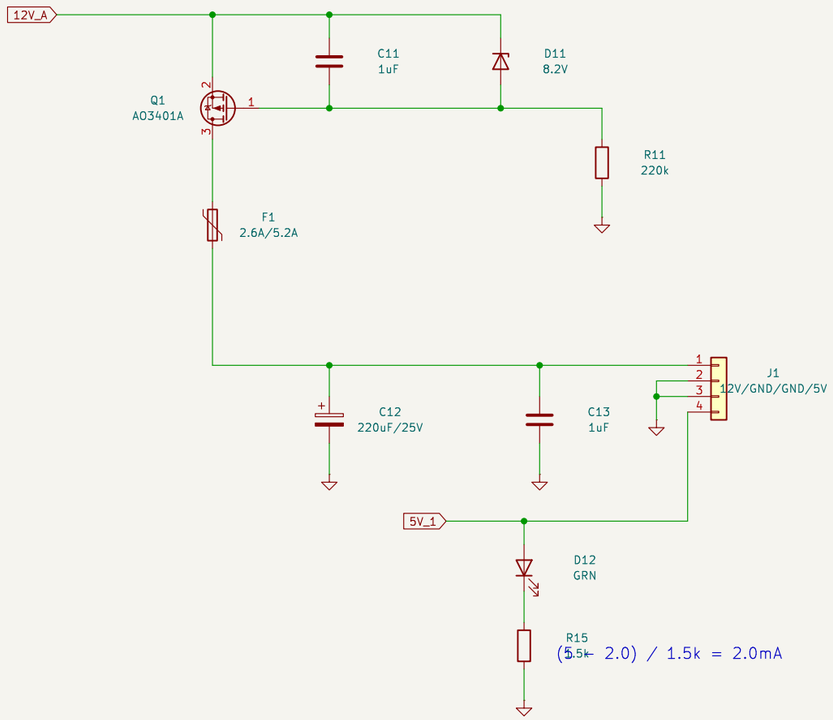
A link is 0R when fitted and simply absent when not. Fitting a link between two populated bucks parallels them, and verify.py check 5 refuses that configuration.



5—Bay SATA Drive PSU
Sheet: /5V rail links/
File: sata_psu_rails.kicad_sch
Title: 5V rail links
Size: A3 | Date: | Rev:
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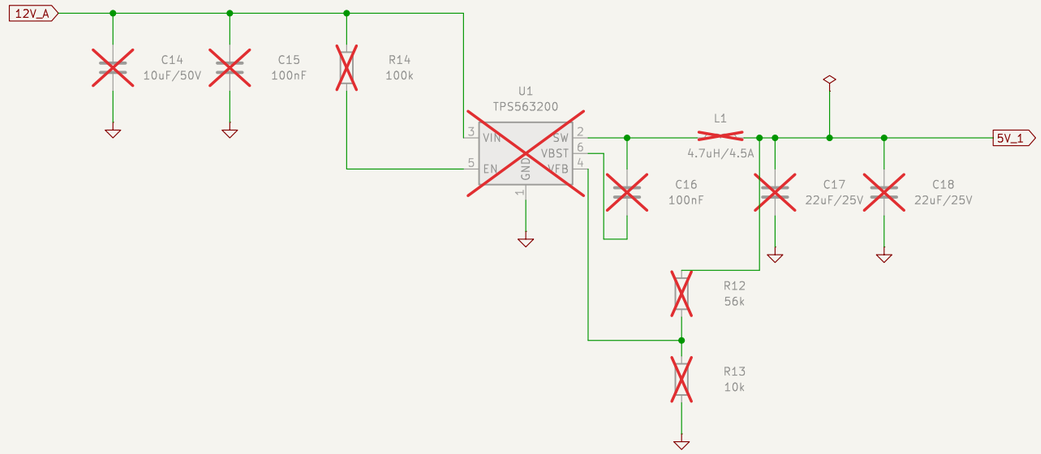
Channel 1 — staggered high-side switch + local 5V

C11=1uF with R11=220k sets turn-on at ~0.2s
D11 clamps Vgs to 8.2V; AO3401A is rated +/-12V



Port 1 local 5V buck [DNP — this port takes 5V from a neighbour]

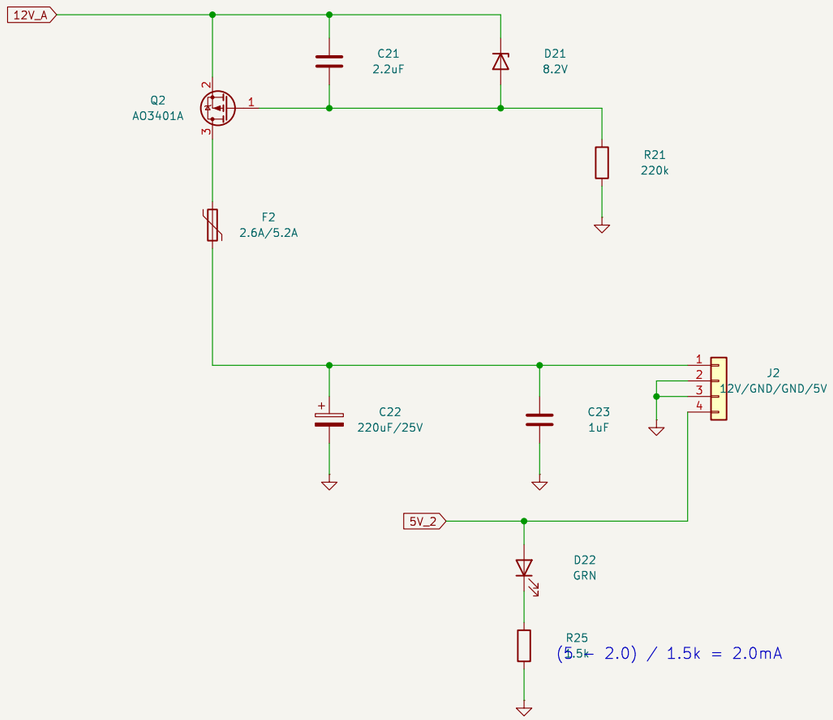
Vout = 0.768V x (1 + R12/R13) = 0.768 x (1 + 56k/10k) = 5.07V
Max load is a 3-port group; see BUCKS/LINKS in gen_sch.py



5-Bay SATA Drive PSU		
Sheet: /Channel 1/		
File: sata_psu_channel1.kicad_sch		
Title: Channel 1		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.3		Id: 4/8

Channel 2 — staggered high-side switch + local 5V

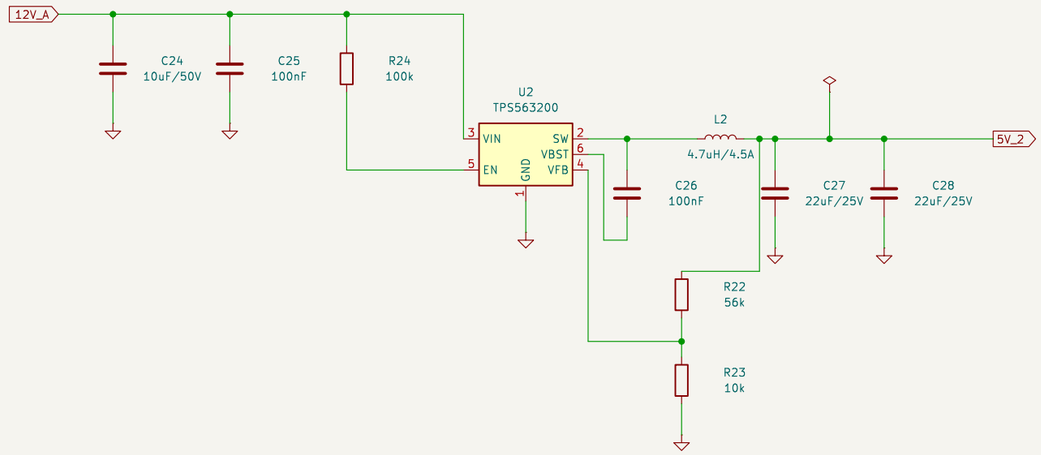
C21=2.2uF with R21=220k sets turn-on at ~0.5s
D21 clamps Vgs to 8.2V; AO3401A is rated +/-12V



(5.5 - 2.0) / 1.5k = 2.0mA

Port 2 local 5V buck

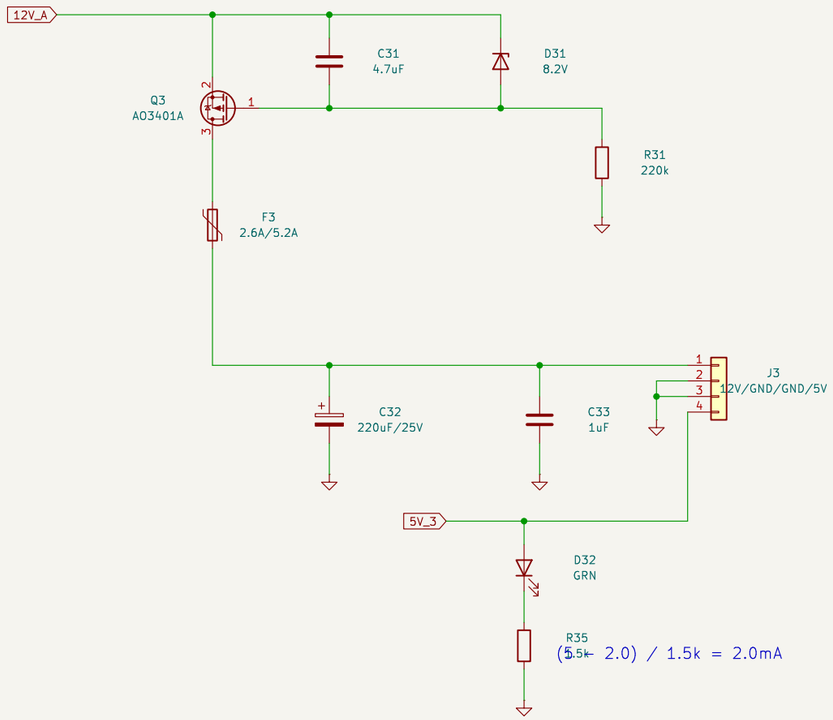
Vout = 0.768V x (1 + R22/R23) = 0.768 x (1 + 56k/10k) = 5.07V
Max load is a 3-port group; see BUCKS/LINKS in gen_sch.py



5-Bay SATA Drive PSU		
Sheet: /Channel 2/		
File: sata_psu_channel2.kicad_sch		
Title: Channel 2		
Size: A3	Date:	**Rev:**
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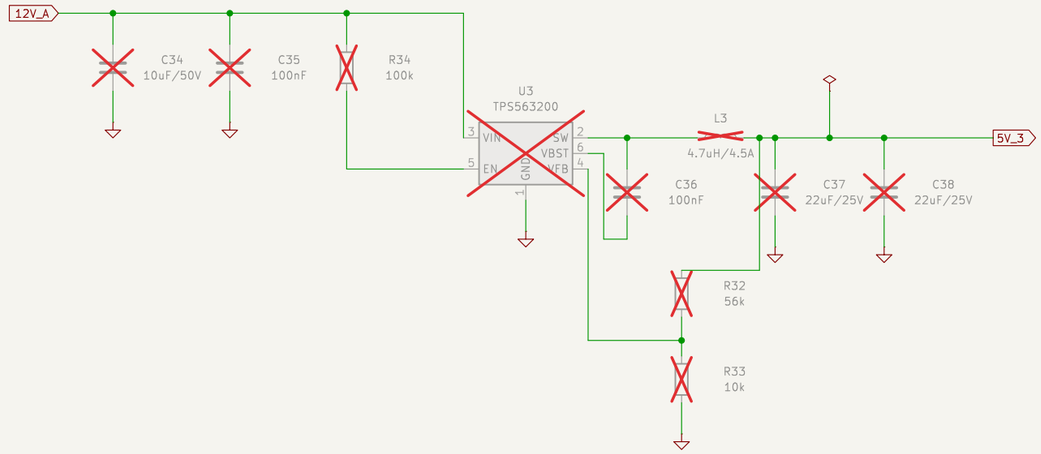
Channel 3 — staggered high-side switch + local 5V

C31=4.7uF with R31=220k sets turn-on at ~1s
D31 clamps Vgs to 8.2V; AO3401A is rated +/-12V



Port 3 local 5V buck [DNP — this port takes 5V from a neighbour]

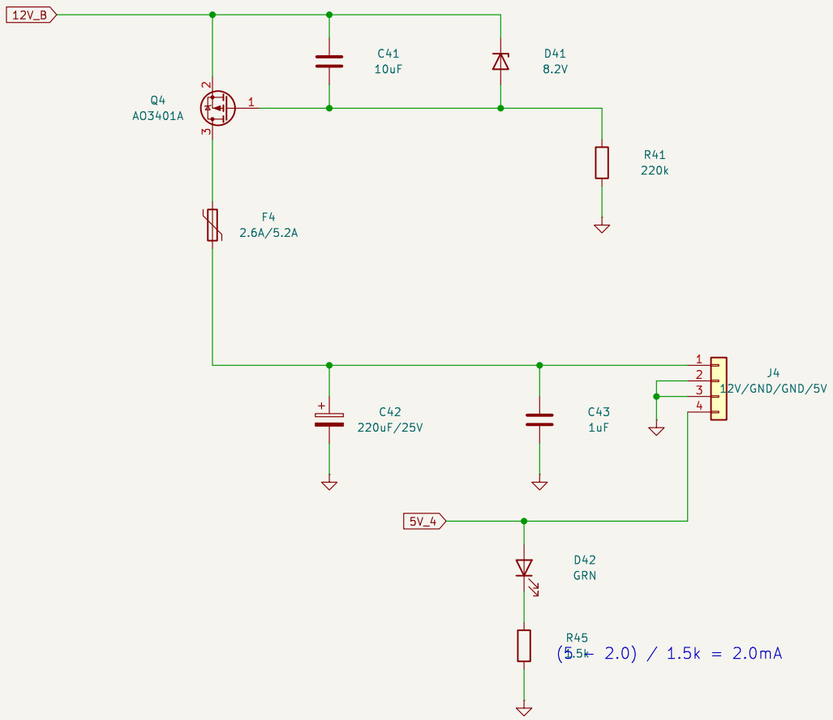
Vout = 0.768V x (1 + R32/R33) = 0.768 x (1 + 56k/10k) = 5.07V
Max load is a 3-port group; see BUCKS/LINKS in gen_sch.py



5-Bay SATA Drive PSU		
Sheet: /Channel 3/		
File: sata_psu_channel3.kicad_sch		
Title: Channel 3		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.3		Id: 6/8

Channel 4 — staggered high-side switch + local 5V

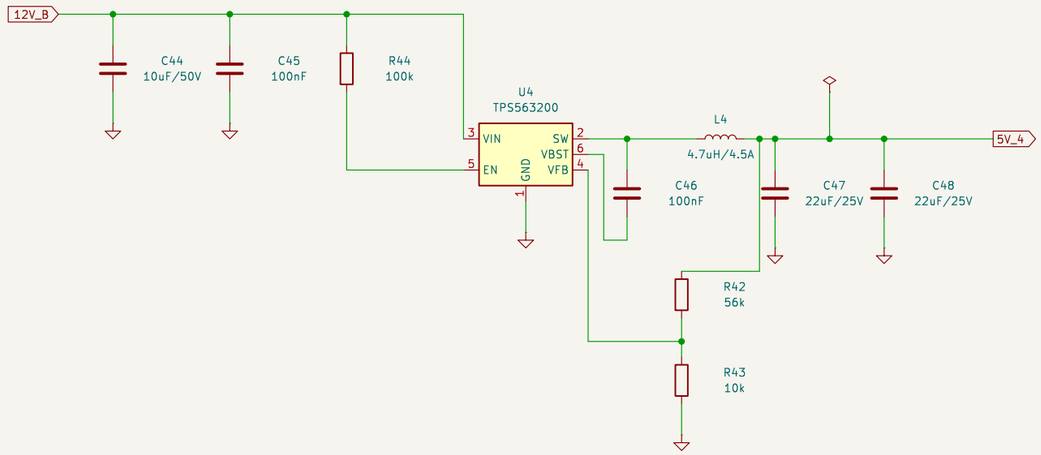
C41=10uF with R41=220k sets turn-on at ~2.2s
D41 clamps Vgs to 8.2V; AO3401A is rated +/-12V



(5.5 - 2.0) / 1.5k = 2.0mA

Port 4 local 5V buck

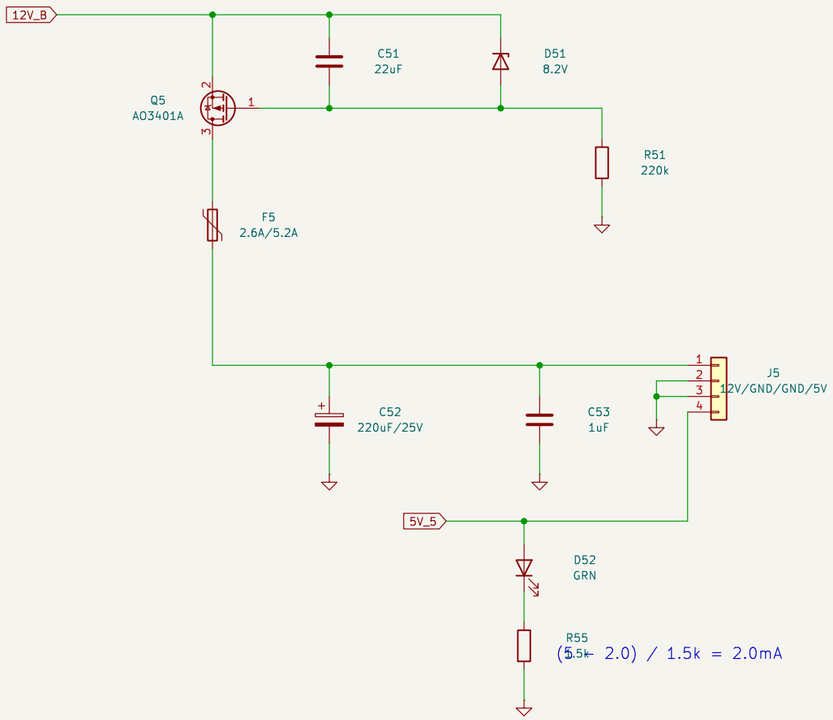
Vout = 0.768V x (1 + R42/R43) = 0.768 x (1 + 56k/10k) = 5.07V
Max load is a 3-port group; see BUCKS/LINKS in gen_sch.py



5-Bay SATA Drive PSU		
Sheet: /Channel 4/		
File: sata_psu_channel4.kicad_sch		
Title: Channel 4		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.3		Id: 7/8

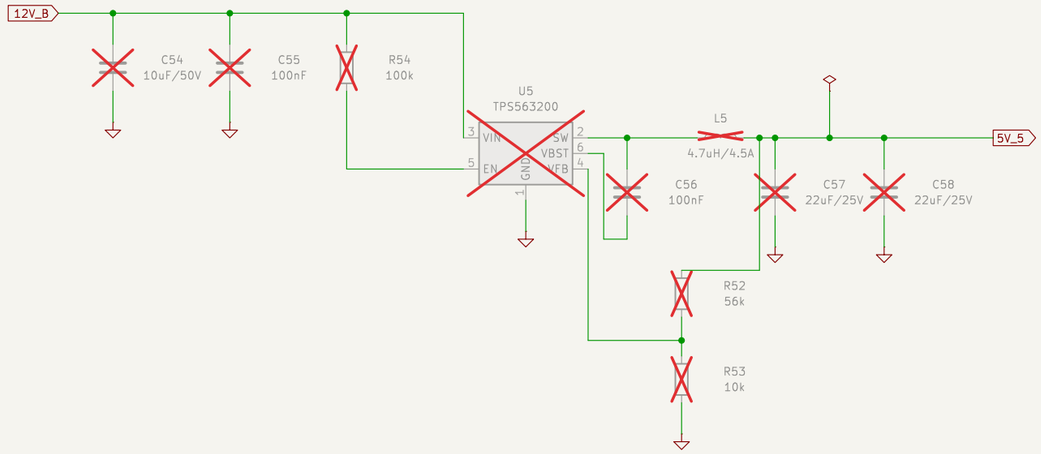
Channel 5 — staggered high-side switch + local 5V

C51=22uF with R51=220k sets turn-on at ~4.8s
D51 clamps Vgs to 8.2V; AO3401A is rated +/-12V



Port 5 local 5V buck [DNP — this port takes 5V from a neighbour]

Vout = 0.768V x (1 + R52/R53) = 0.768 x (1 + 56k/10k) = 5.07V
Max load is a 3-port group; see BUCKS/LINKS in gen_sch.py



5-Bay SATA Drive PSU

Sheet: /Channel 5/
File: sata_psu_channel5.kicad_sch

Title: Channel 5

Size: A3	Date:	**Rev:**
KiCad E.D.A. 9.0.3		Id: 8/8